

Wireless LANs

Design goals for wireless LANs

- Low power for battery use
- Global, seamless operation
- No special permissions or licenses needed to use the WLAN
- Robust transmission technology
- Easy to use for everyone, simple management
- Protection of investment in wired networks
- Security (no one should be able to read my data), privacy (no one should be able to collect user profiles), safety (low radiation).

Wireless LAN Requirements

- **Throughput** - more work completed per unit time
- **Number of nodes** - hundreds across cells
- **Service area** - 100+ meters
- **Battery power consumption** - sleep when not in use
- **Transmission robustness and security** - reliable transmission, and maintain security
- **Network operation** - minimize interference between neighboring networks
- **License-free operation** - better to operate without licensed frequencies
- **Handoff/roaming** - MAC protocol should support smooth handoffs
- **Dynamic configuration** - MAC addressing should support automatic addition and deletion of addresses

IEEE 802 Standards

- 802.1: High Level Interface
- 802.2: Logical Link Control
- 802.3: CSMA/CD Networks
- 802.4: Token Bus Networks
- 802.5: Token Ring Networks
- 802.6: Metropolitan Area Networks
- 802.7: Broadband Networks
- 802.8: Fiber Optic Networks
- 802.9: Integrated Data and Voice Networks
- 802.10 Virtual LANs
- 802.11 Wireless LANs
- 802.12 Communication media
- 802.14 Data transport over traditional cable TV network
- 802.15; personal area networks
- 802.16 Wireless Local Loops

LAN Stack

OSI Stack

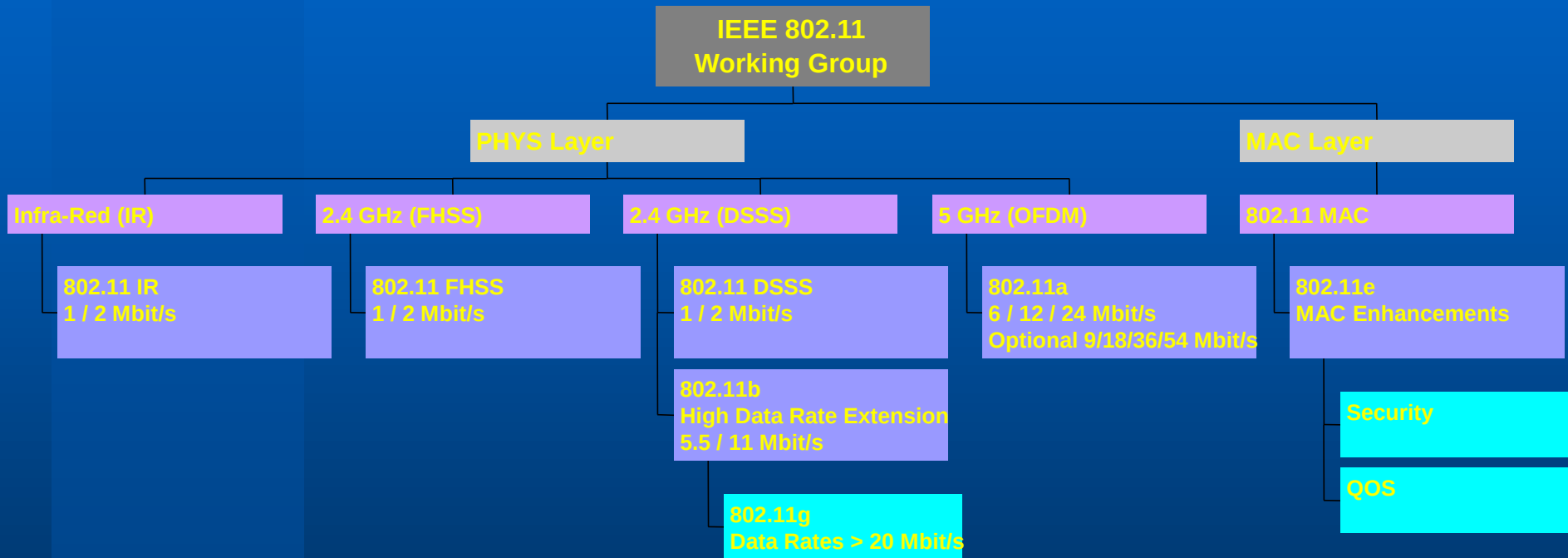
LAN Stack



Physical Layer

2.4 GHz frequency hopping spread spectrum 1 Mbps 2 Mbps	2.4 GHz direct sequence spread spectrum 1 Mbps 2 Mbps	Infrared omni- directional 1 Mbps 2 Mbps	5-GHz Orthogonal FDM 6,9,12 18,24,36, 48, 54 Mbps	2.4 GHz direct sequence spread spectrum 5.5 Mbps 11 Mbps	2.4 GHz Orthogonal FDM Up to 54 Mbps
IEEE 802.11			IEEE 802.11a	802.11b	802.11g

IEEE 802.11 Organization Tree:



Source IEEE

802.11a/b/g

Operates in 2 different modes:

– Infrastructure mode

- Associates with an access point
- All communication goes through the access point
- Used for wireless access at a company or campus

– Peer-to-Peer Ad Hoc Mode

- If two nodes are within range of each other they can communicate directly with no access point
- A few users in a room could quickly exchange files with no access point required

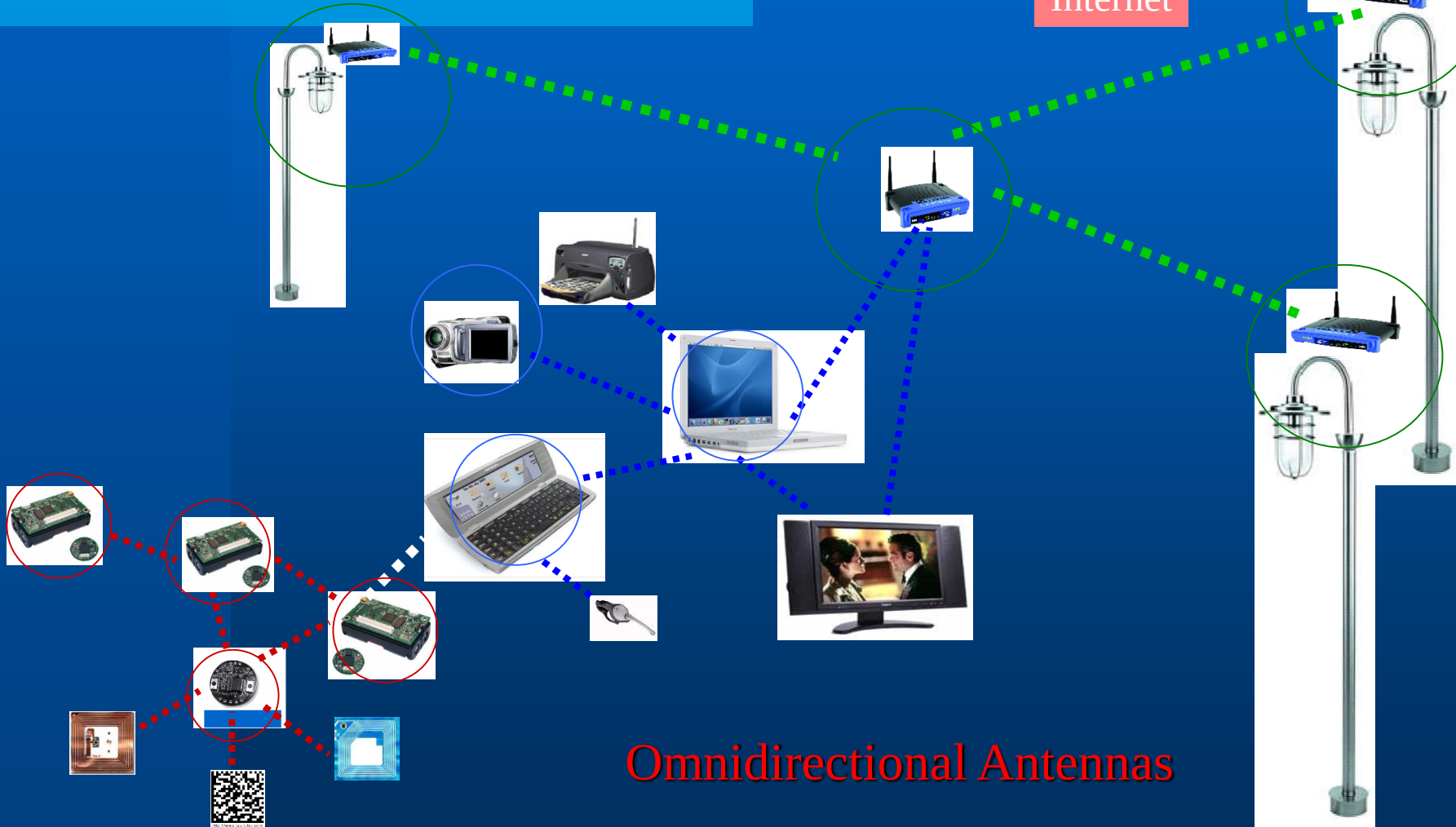
IEEE 802.11 Wireless LAN

- **802.11b**
 - 2.4-5 GHz unlicensed radio spectrum
 - up to 11 Mbps
 - direct sequence spread spectrum (DSSS) in physical layer
 - all hosts use same chipping code
 - widely deployed, using base stations
- **802.11a**
 - 5-6 GHz range
 - up to 54 Mbps
- **802.11g**
 - 2.4-5 GHz range
 - up to 54 Mbps
- All use CSMA/CA for multiple access
- All have base-station and ad-hoc network versions

Infrastructure, Ad-hoc, Mesh, Sensors Networks



Internet

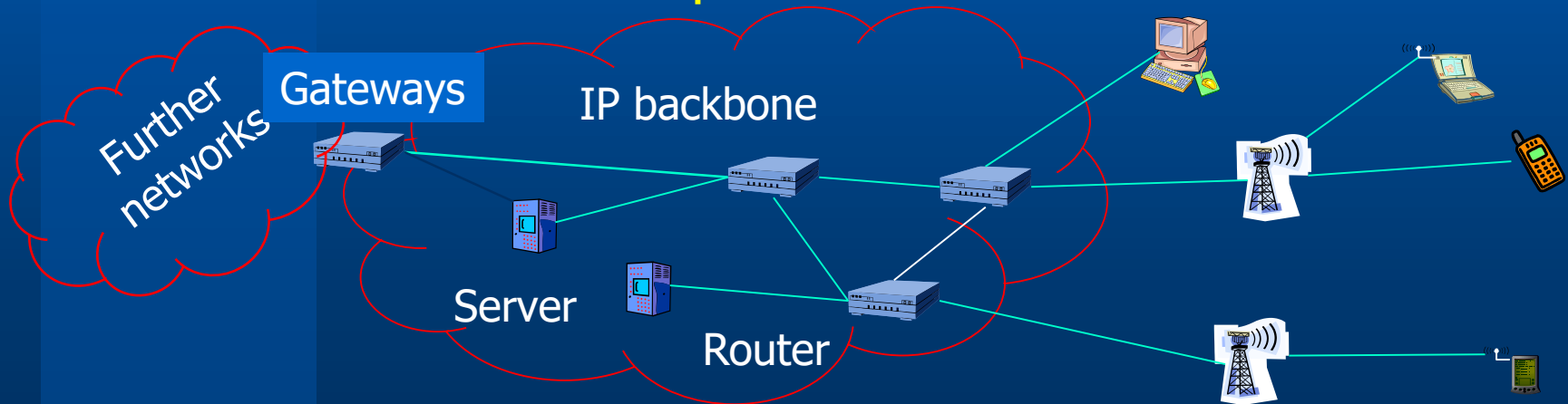


Omnidirectional Antennas

Infrastructure-based Wireless Networks

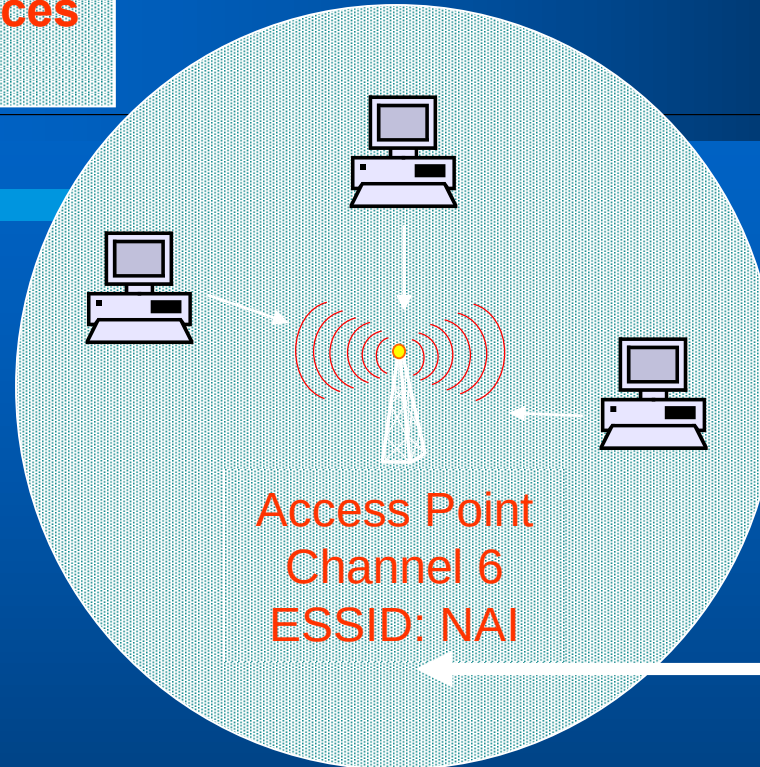
- Typical wireless network are based on infrastructure

- E.g., GSM, UMTS, WLAN, ...
- Base stations connected to a wired backbone network
- Mobile entities communicate wirelessly to these base stations
- Traffic between different mobile entities is relayed by base stations and wired backbone
- Mobility is supported by switching from one base station to another
- Backbone infrastructure required for administrative tasks



Wireless Cells

11 Mbps bandwidth
“shared” by all devices
in the Cell!

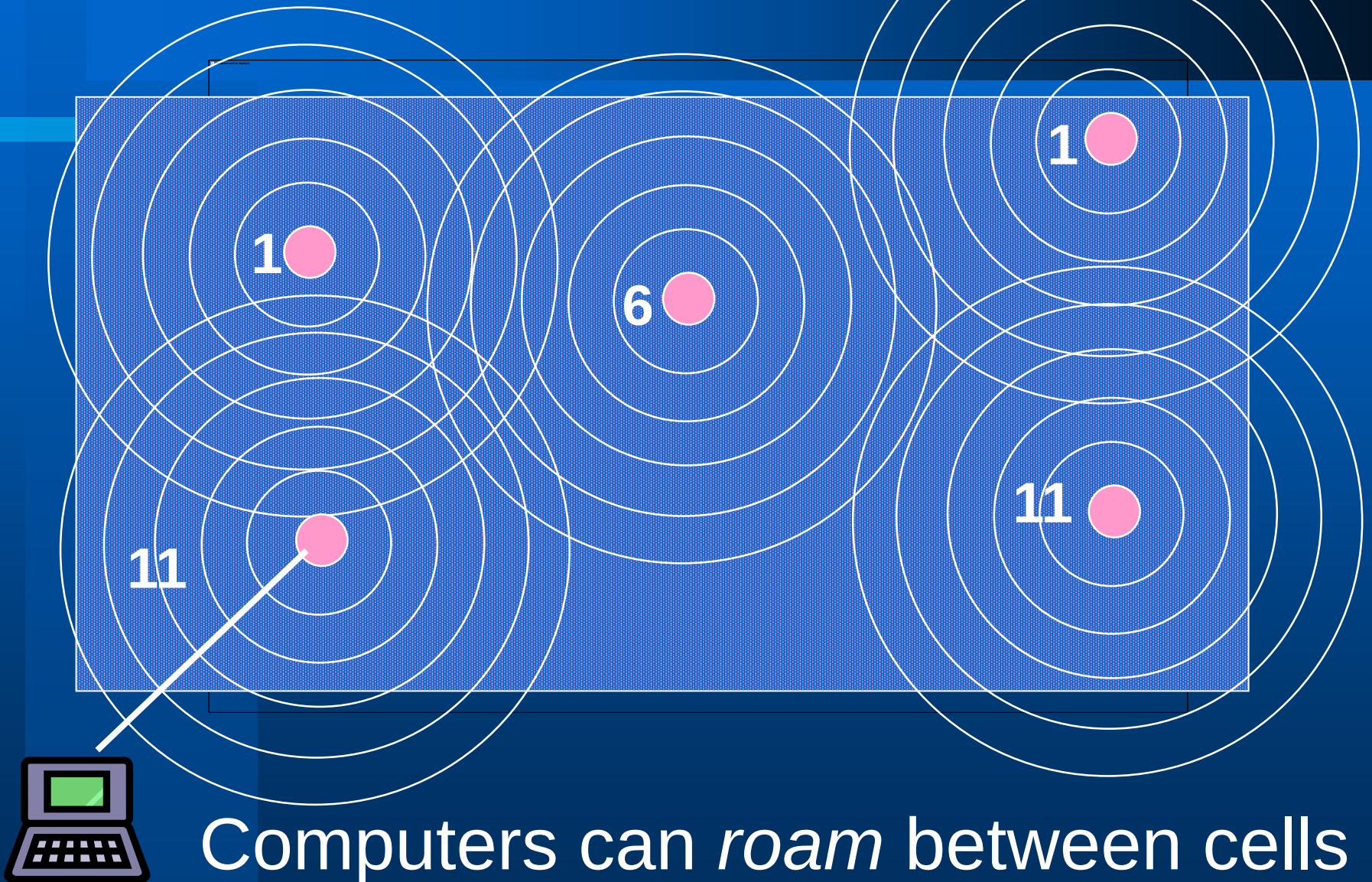


Access Point
coverage
area is called a
“Cell”

Range per Access
Point is 100m

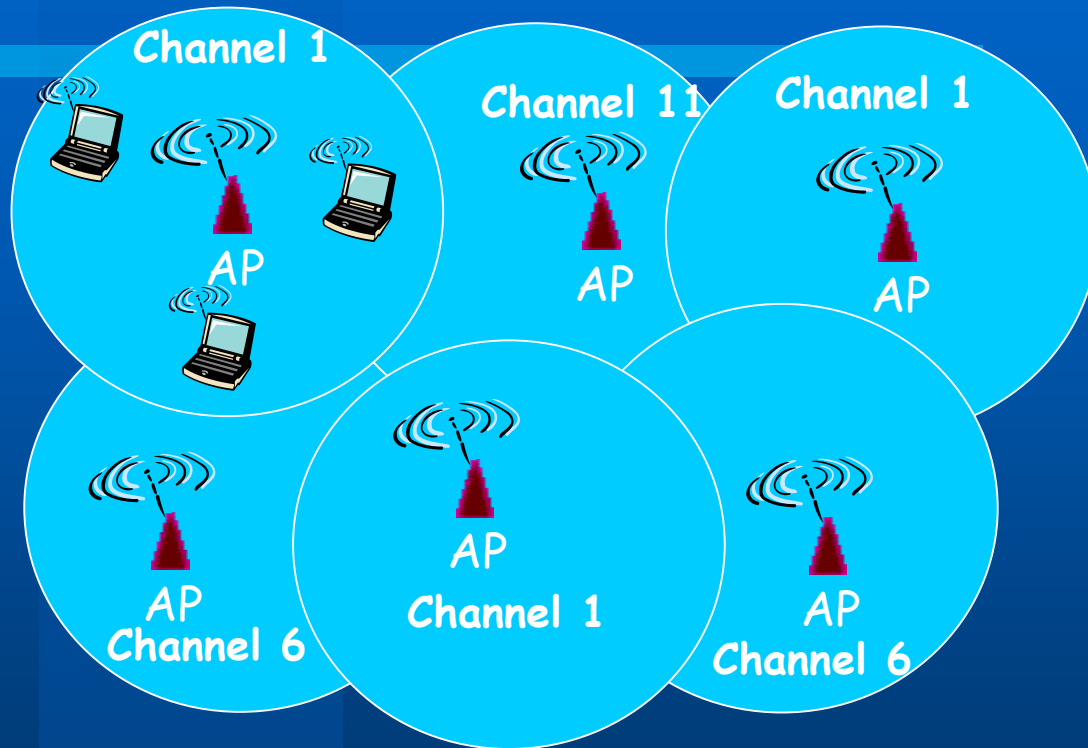
- In Canada/US, there are eleven 802.11 channels
- Only channels 1, 6 and 11 are non-overlapping
- Each Access Point coverage area is called a “Cell”
- Computers can *roam* between cells

Wireless Cells



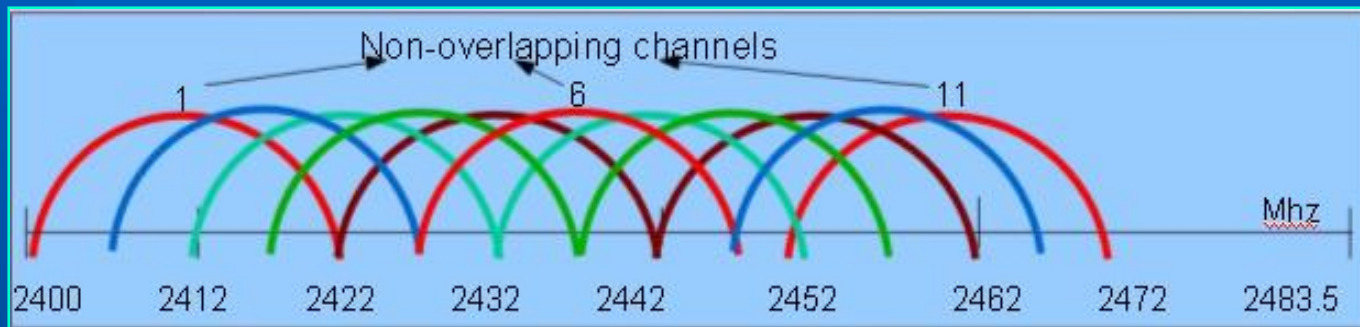
Computers can *roam* between cells

Wireless Cells



- 802.11 has 11 channels
- Channels 1, 6, and 11 are non-overlapping.
- Each AP coverage area is called a “cell”
- Wireless nodes can roam between cells

802.11b Channels



- Each channel is 22MHz wide
- Adjacent channels overlap
- Non-overlapping channels: 1, 6, 11.

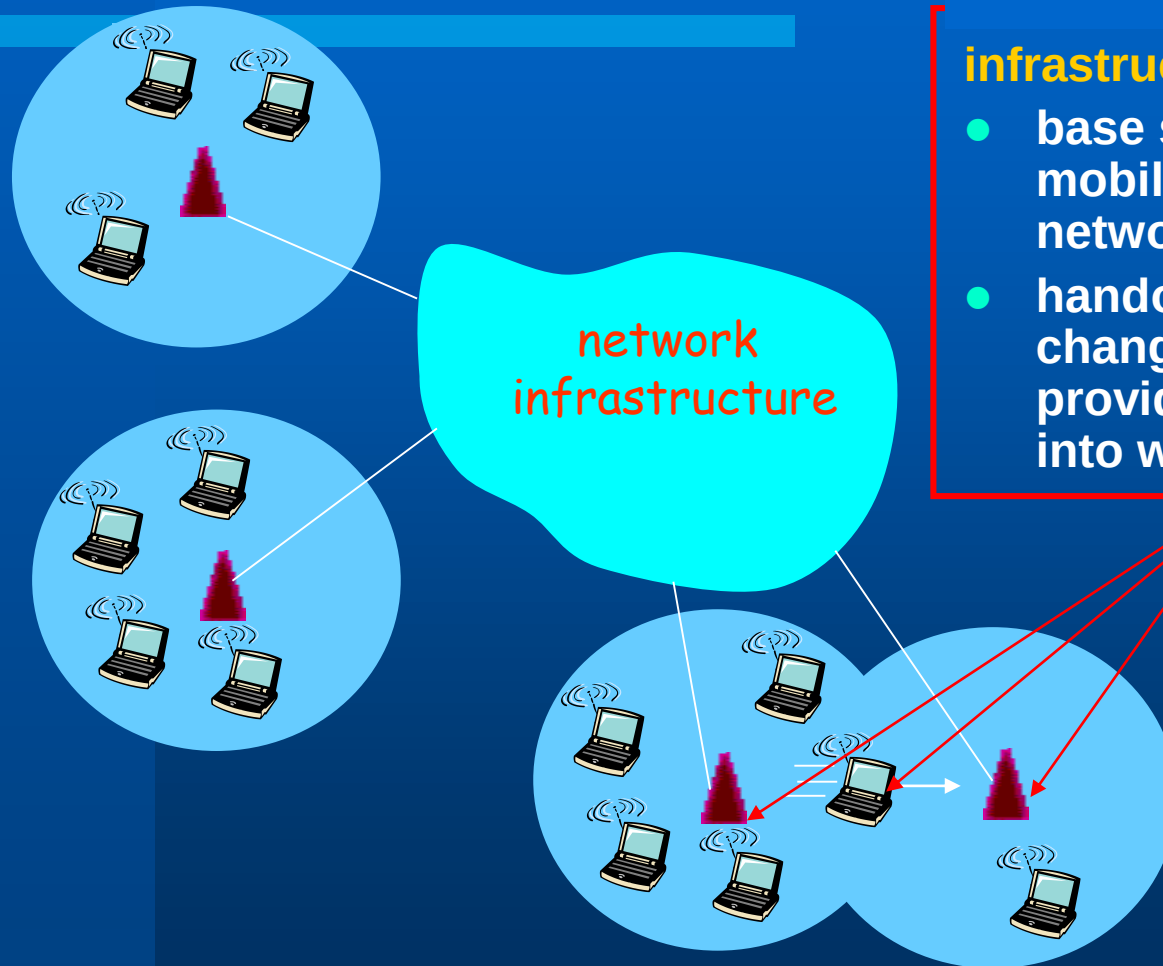
Infrastructure Access

- Access Points:

- Provide infrastructure access to mobile users
- Cover a fixed area
- Wired into LAN



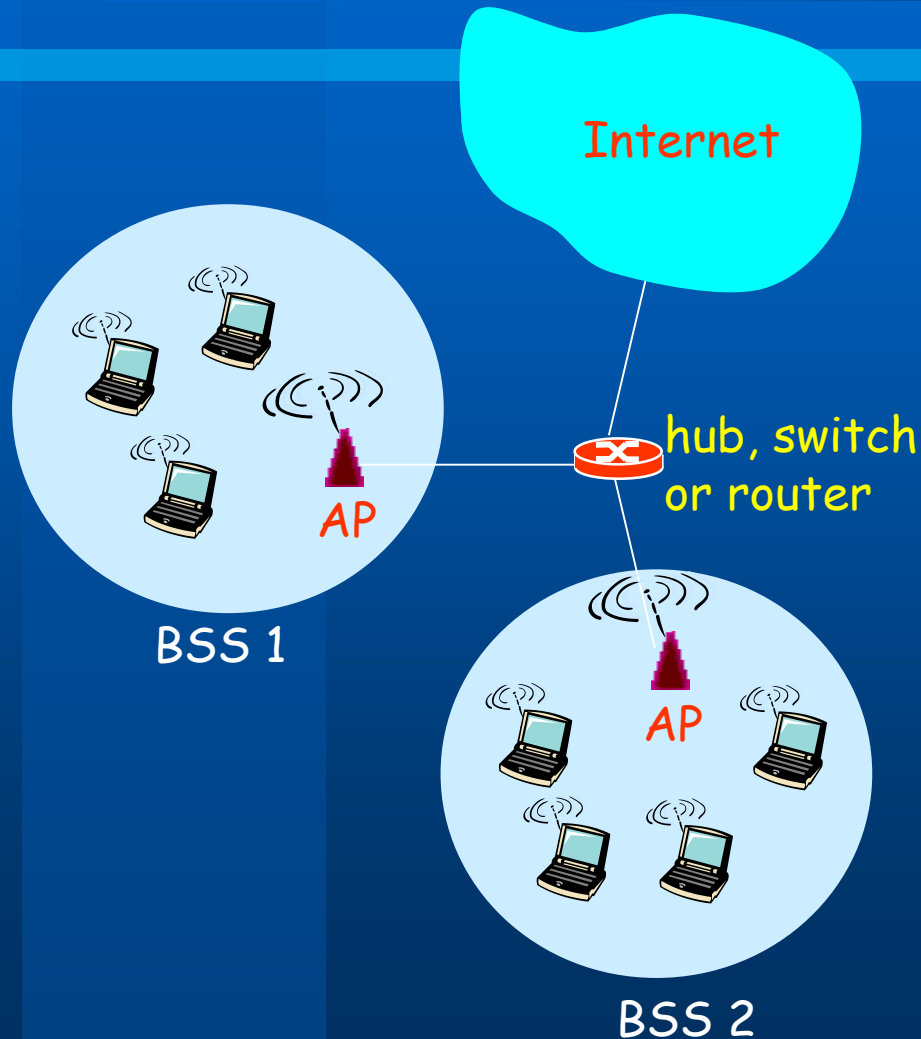
Infrastructure Mode



infrastructure mode

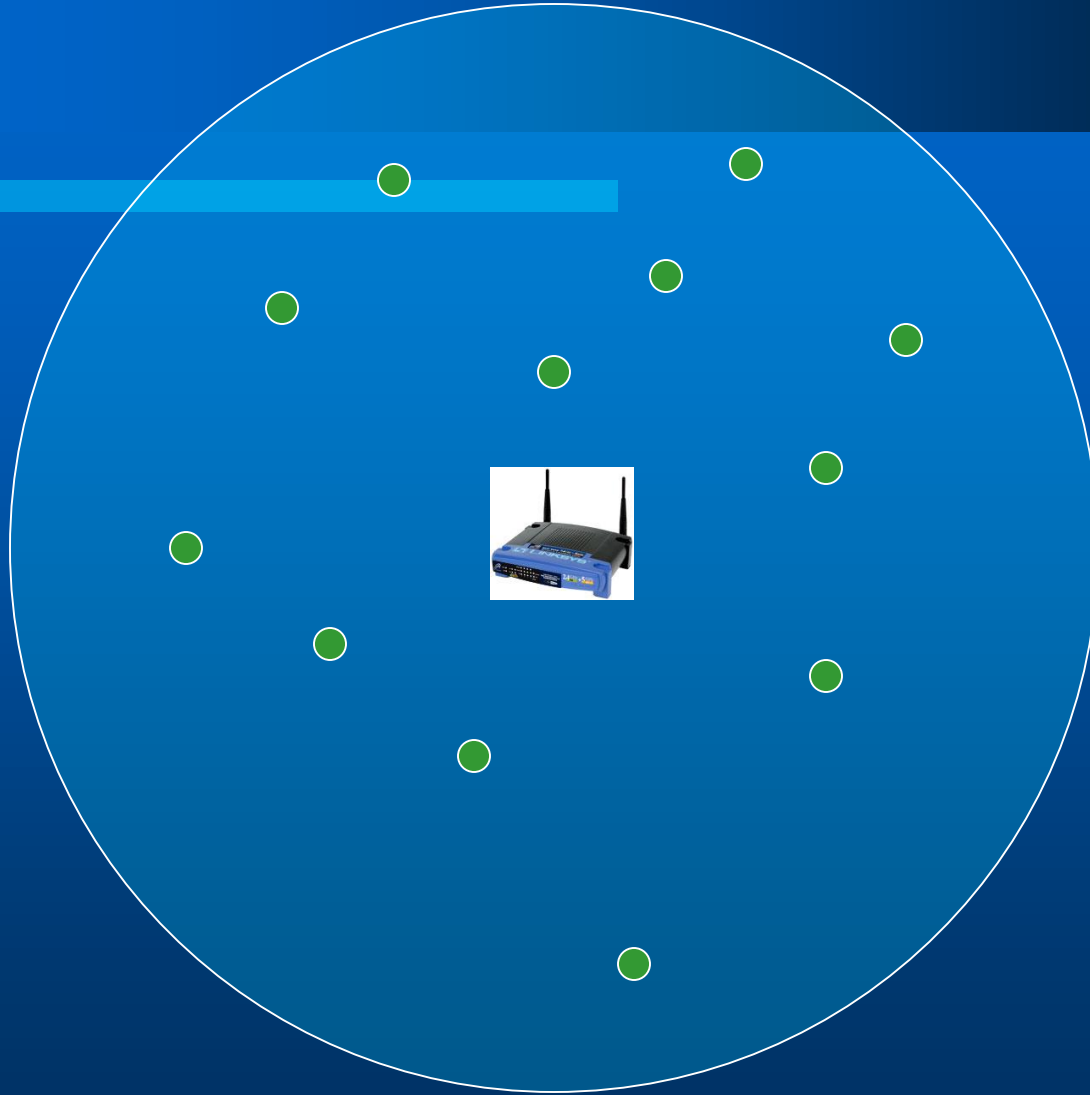
- base station connects mobiles into wired network
- handoff: mobile changes base station providing connection into wired network

802.11 LAN architecture

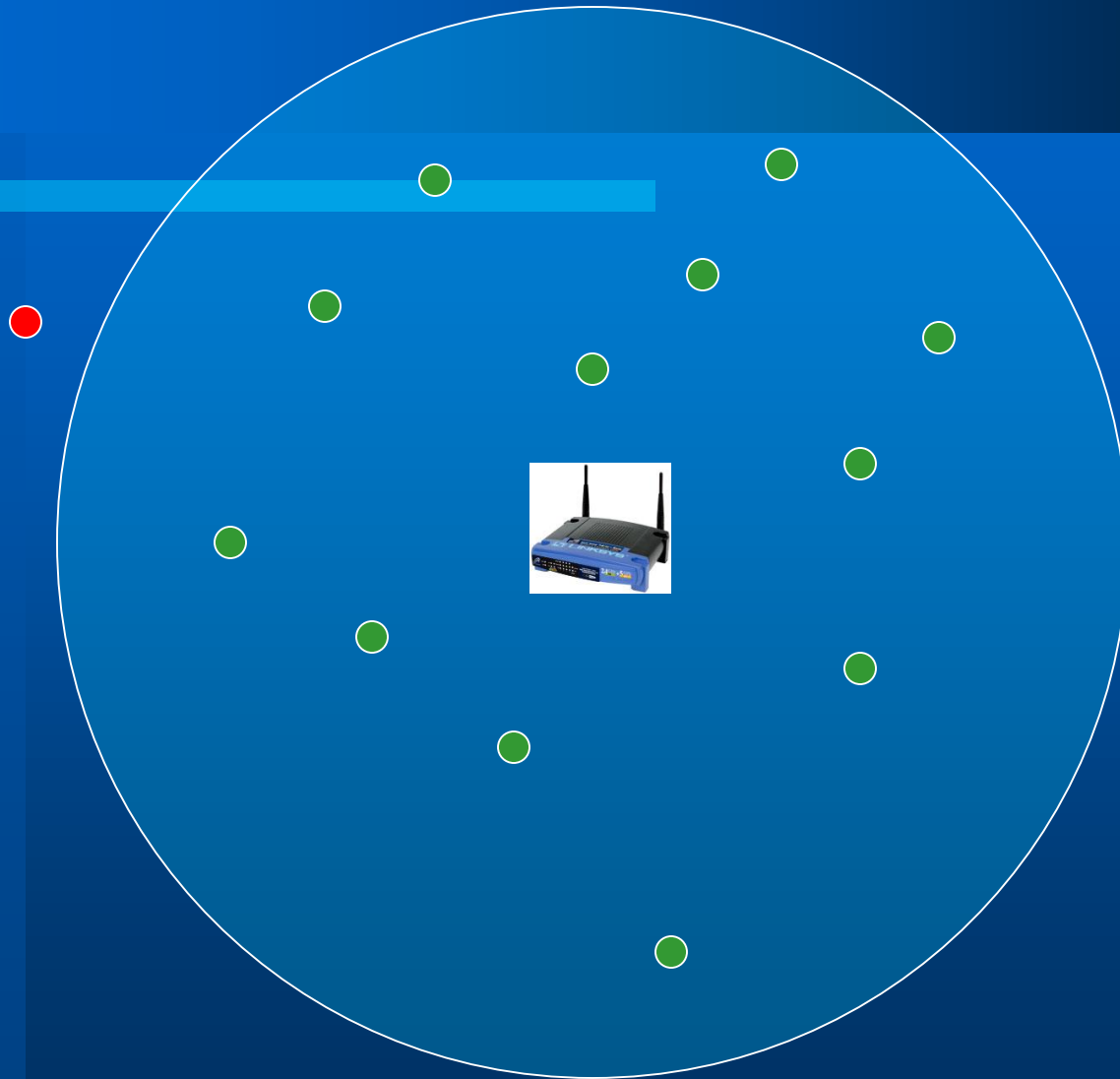


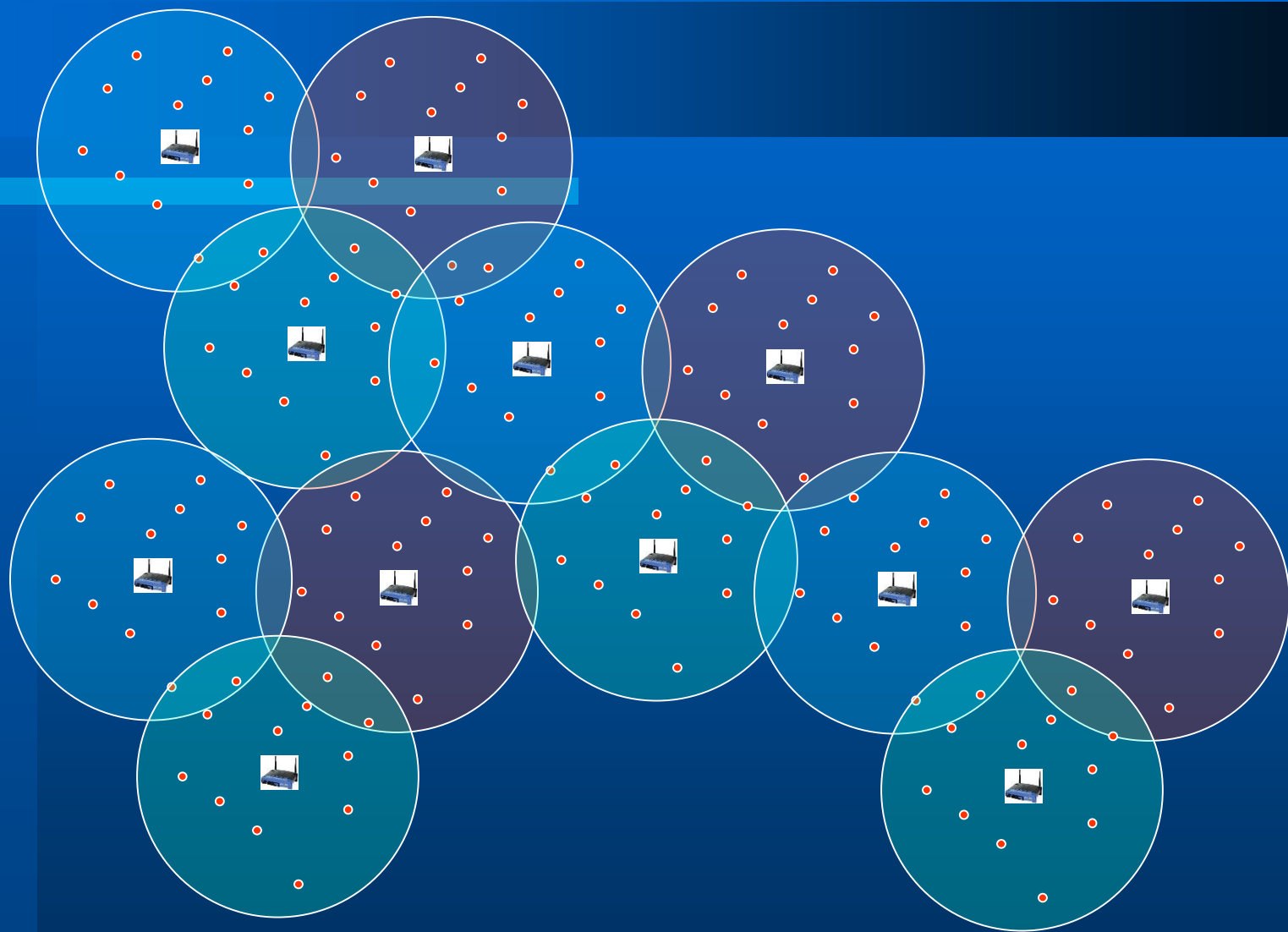
- **wireless host** communicates with base station
 - **base station = access point (AP)**
- **Basic Service Set (BSS)** “cell” infrastructure mode contains:
 - **wireless hosts**
 - **access point (AP): base station**
 - **ad hoc mode: hosts only**

Infrastructure Access

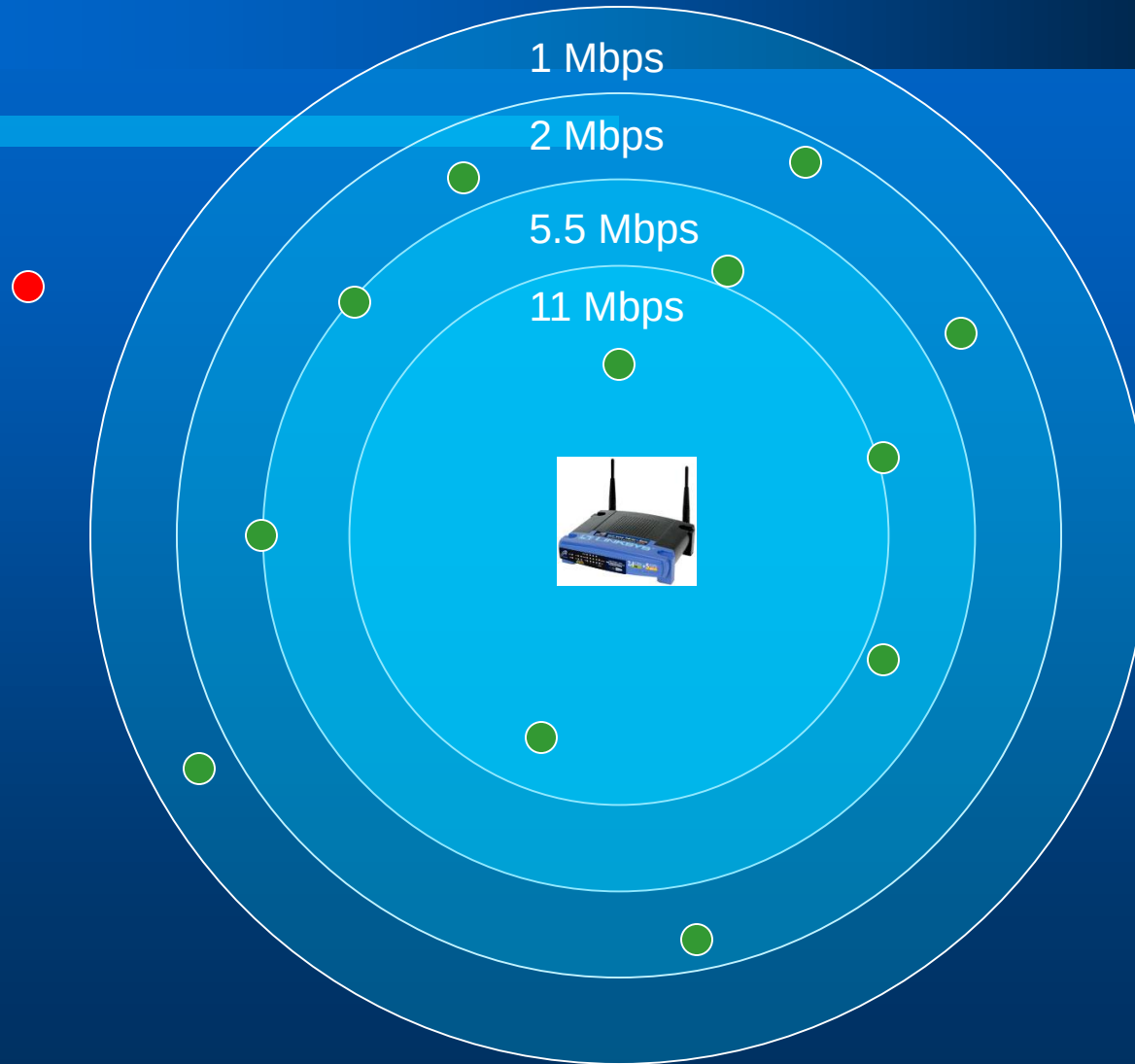


Infrastructure Access





802.11a/b/g are multi-rate devices



Features

- **Power Management**

- mobile nodes can “sleep” to save power
- AP will buffer frames until client requests them
- AP can use virtual bitmap field in beacons to indicate which stations have data waiting

- **Security**

- Wired Equivalent Privacy (WEP)

Summary

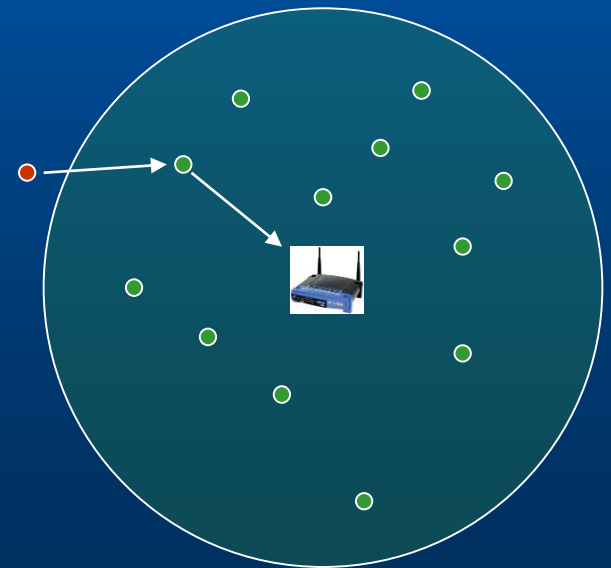
- IEEE 802.11b (WiFi) is a wireless LAN technology that is rapidly growing in popularity.
- Convenient, inexpensive, easy to use.
- Growing number of “hot spots” everywhere
 - airports, hotels, bookstores, institutes etc
- Estimates: 70% of WLANs are insecure!

Problems

- Access Point placement depends on wired network availability
- Obstructions make it difficult to provide total coverage of an area
- Site surveys are performed to determine coverage areas
- Security Concerns: rogue access points in companies etc..
- Each Access Point has limited range

Multi-hop Infrastructure Access

- Nodes might be out of range of the access point, BUT in range of other nodes.
- The nodes in range of the access point could relay packets to allow out of range nodes to communicate.
- NOT part of 802.11



How can this be done?

- **ROUTING!!**

- **Wired Networks:**

- **Hierarchical Routing**

- Network is divided into subnets
 - Nodes look at netmask and determine if the address is directly reachable. If not, just forward to the default gateway.
 - Different protocols for different levels of the hierarchy
 - RIP, OSPF, BGP

Wireless Routing

- Flat routing
 - You can't assume that since a node is in your subnet that it is directly accessible
 - Node must maintain or discover routes to the destination
 - All nodes are routers

Mobile Ad Hoc Networks (MANET)

Definition

- An ad hoc network is a transitory association of mobile nodes which do not depend upon any fixed support infrastructure.
- Connection and disconnection is controlled by the distance among nodes.

MANETs - General Introduction

- Formed by wireless nodes which may be mobile.
- Without (necessarily) using a pre-existing infrastructure.
- All or most wireless nodes are hosts as well as routers
- Routes between nodes may potentially contain multiple communication hops.
- Highly dynamic network topology undergoing frequent changes.
- All or most nodes have limited power resources due to battery based operation.

MANETs - General Introduction

Advantages of Mobile Ad-Hoc Networks

- Ease and speed of deployment.
- Decreased or no dependence on infrastructure

Ad-Hoc Network Applications

- Personal area networking
- Military environments

Civilian environments

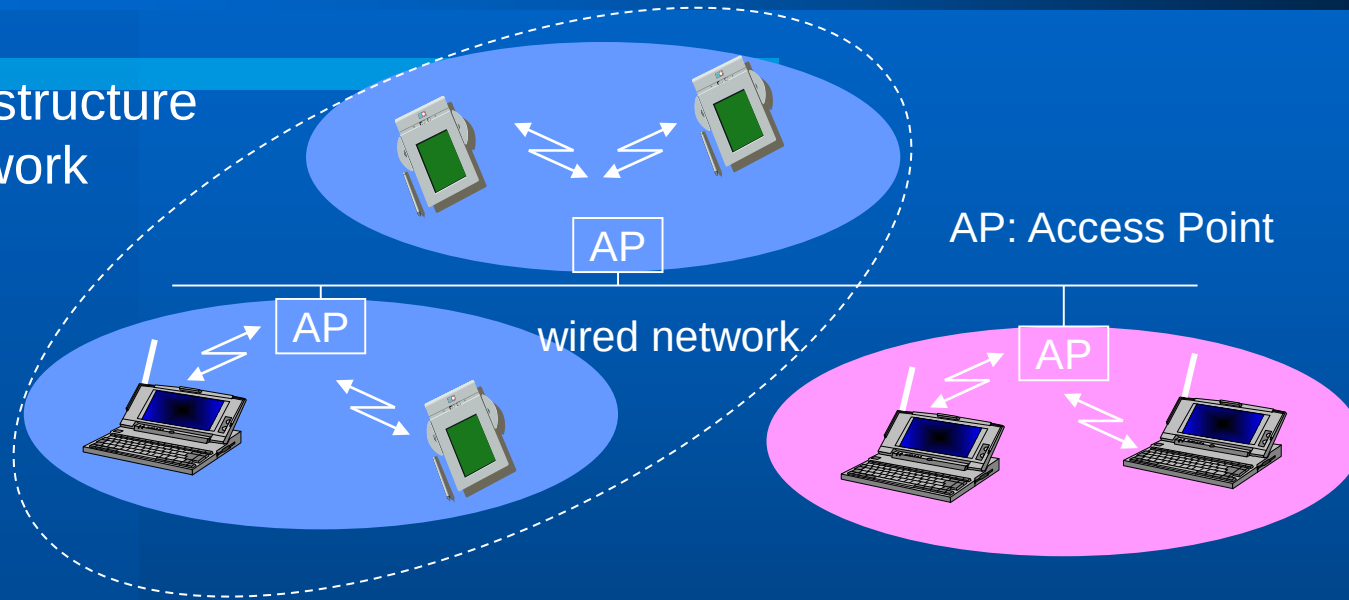
- Taxi network
- Meetings and social gatherings
- Sports stadiums
- Boats, Aircraft

Emergency operations

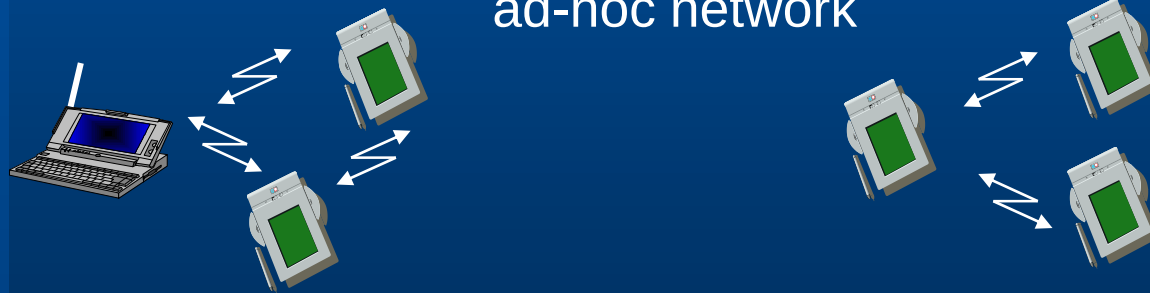
- Search-and-rescue
- Policing and fire fighting

Infrastructure vs Adhoc

infrastructure
network

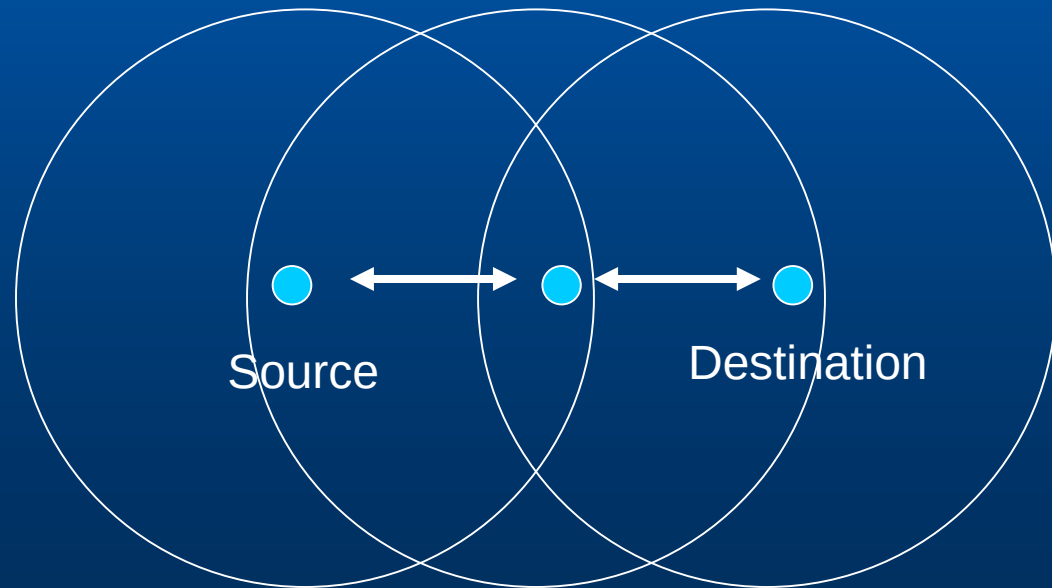


ad-hoc network



Multi-hop Ad Hoc Network

- If communication is required between two nodes which are out of range of each other, intermediary nodes can forward the packets.

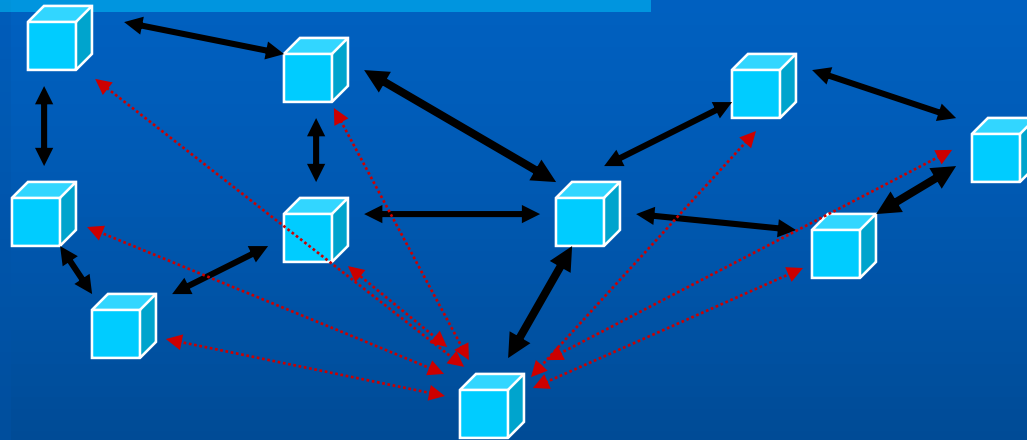


Wireless Routing

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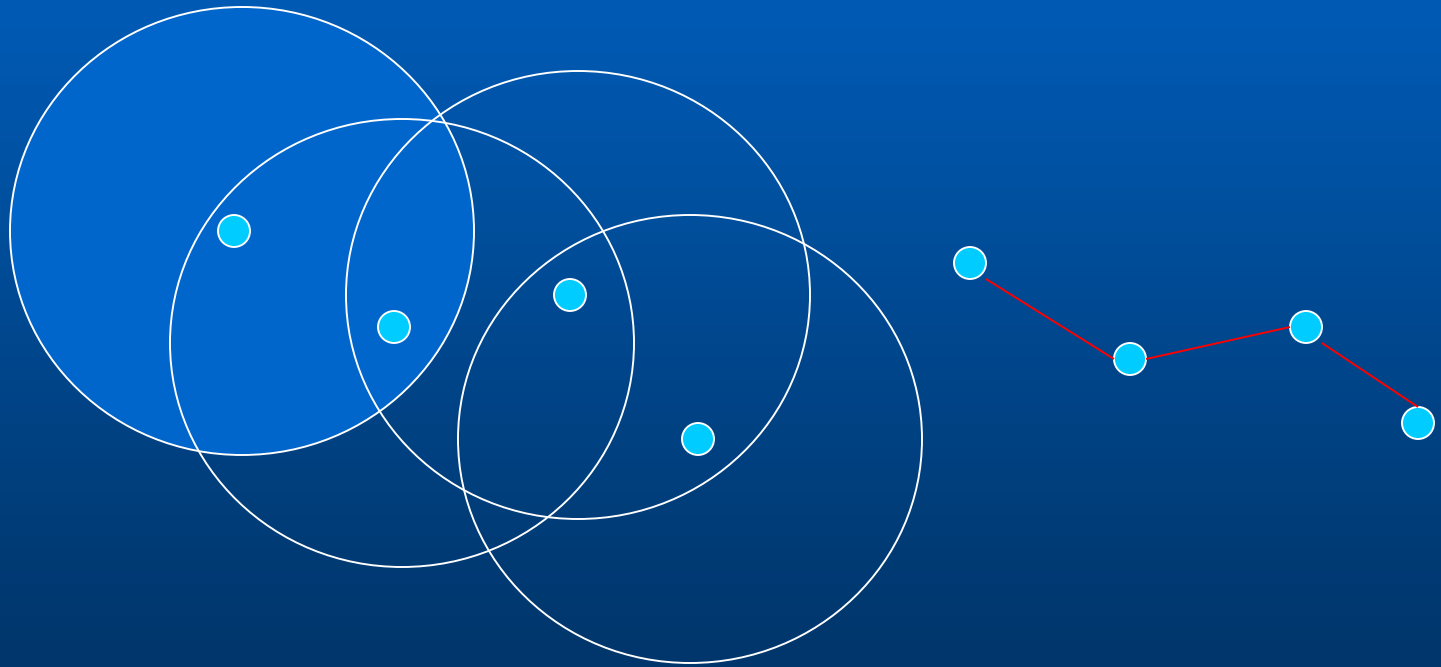
Ad-Hoc Networks



- Peer-to-peer communications.
- No backbone infrastructure.
- Routing can be multihop.
- Topology is dynamic.
- Fully connected with different links.

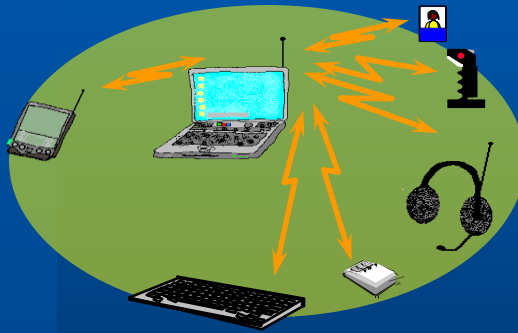
Multi-Hop Wireless - Mobility Mobile Ad Hoc Networks (MANET)

- **Mobility causes route changes**

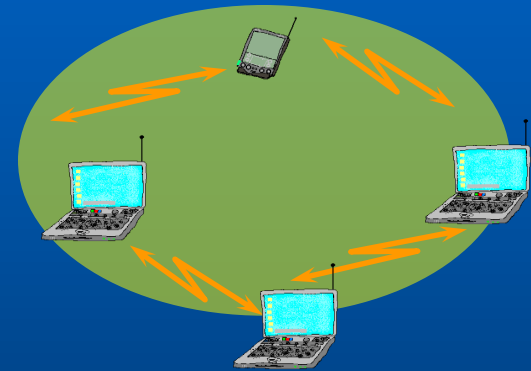


Usage Scenarios

- Cable replacement
- Adhoc PAN



Cable Replacement



Personal Ad-hoc Networks

Peer to Peer Ad Hoc Mode



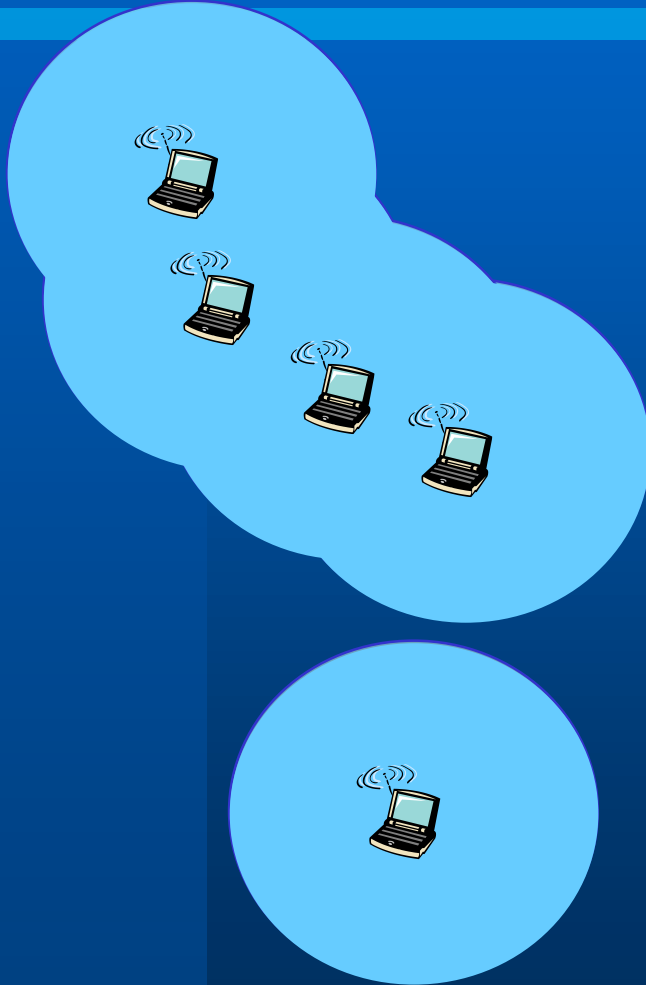
Design Issues

- Ad-hoc networks provide a flexible network infrastructure for many emerging applications.
- The capacity of such networks is generally unknown.
- Transmission, access, and routing strategies for ad-hoc networks are generally ad-hoc.
- Cross layer design critical and very challenging.
- Energy constraints impose interesting design tradeoffs for communication and networking.

Battery-operated Devices: Energy-efficient Operation

- Often (not always!), participants in an ad hoc network draw energy from batteries
- Desirable: long run time for
 - Individual devices
 - Network as a whole
- Energy-efficient networking protocols
 - E.g., use multi-hop routes with low energy consumption (energy/bit)
 - E.g., take available battery capacity of devices into account
 - How to resolve conflicts between different optimizations?

Ad hoc Mode



Ad hoc mode

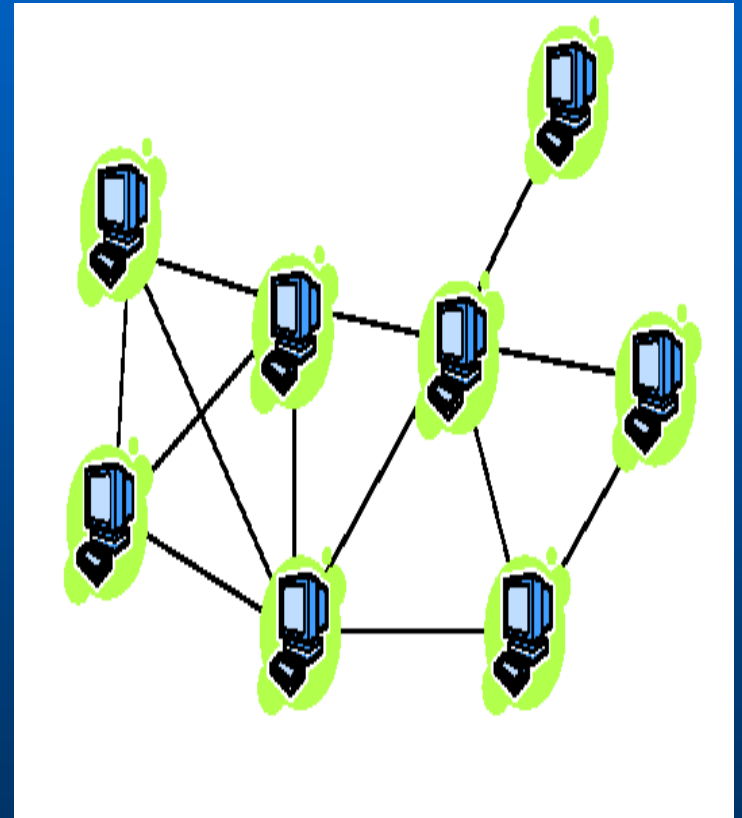
- no base stations
- nodes can only transmit to other nodes within link coverage
- nodes organize themselves into a network: route among themselves

Ad-hoc networks

- Each node is willing to forward data
- NO dedicated routing hardware
 - flexible to node/link failures

Ad-hoc networks provide a flexible network infrastructure for many emerging applications.

The capacity of such networks is generally unknown.



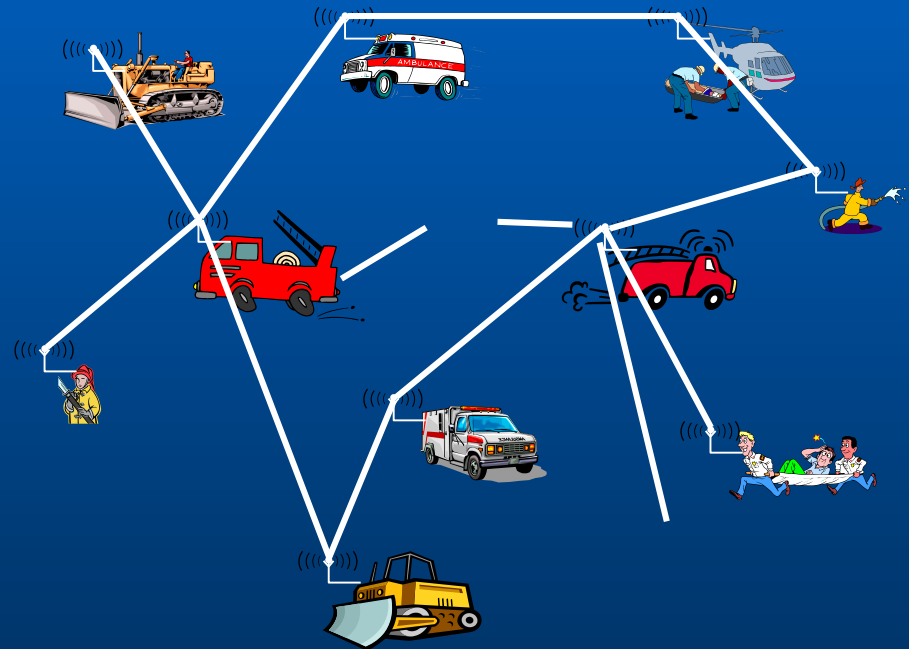
Ad Hoc Networks vs. Infrastructure-based Networks

	Infrastructure-based network	Ad hoc network
Prerequisites	Pre-deployed infrastructure, e.g. routers, switches, base stations, servers	None
Node properties	End system only	Duality of end system and network functions
Connections	Wired or wireless	Usually wireless
Topology	Outlined by the pre-deployed infrastructure	Self-organized topology maintained by the nodes
Network functions	Provided by the infrastructure	Distributed to all participating nodes

Mobility: Suitable, Adaptive Protocols

- In many (not all!) ad hoc network applications, participants move around
 - In cellular network: simply hand over to another base station

- ❑ In *mobile ad hoc networks (MANET)*:
 - ❑ Mobility changes neighborhood relationship
 - ❑ Must be compensated for
 - ❑ E.g., routes in the network have to be changed
- ❑ Complicated by scale
 - ❑ Large number of such nodes difficult to support



Challenges

- Limited wireless transmission range
- Broadcast nature of the wireless medium
- Hidden terminal problem
- Packet losses due to transmission errors
- Mobility-induced route changes
- Mobility-induced packet losses
- Battery constraints

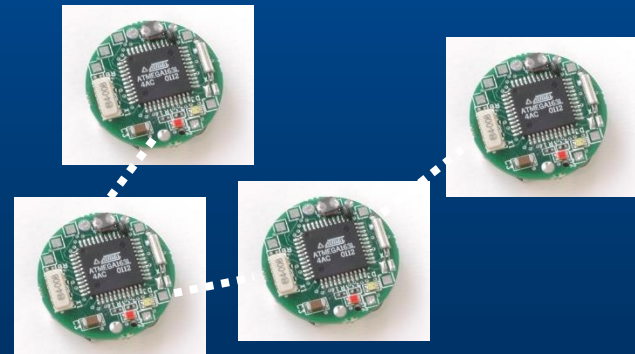
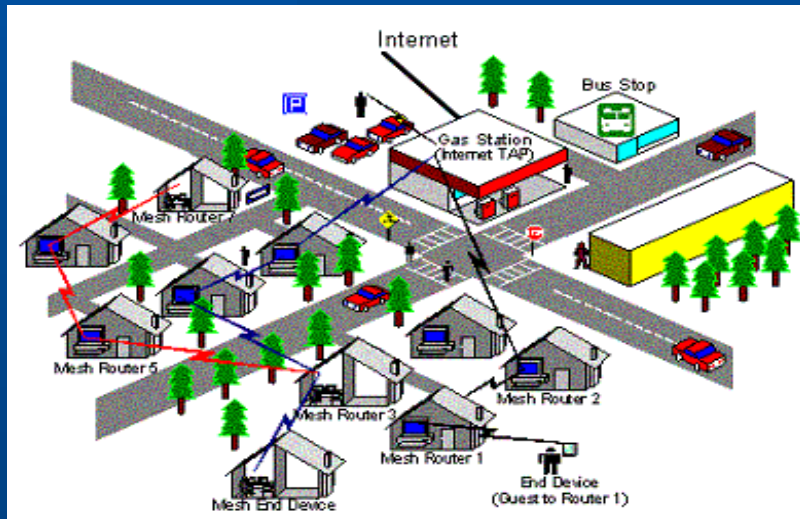
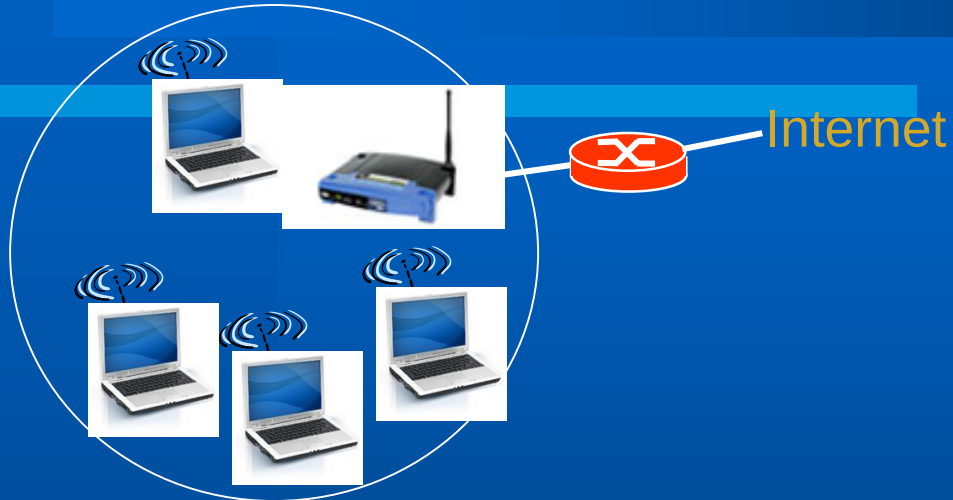
Problems

- Communication is only possible between nodes which are directly in range of each other.

Problems for both Infrastructure and Ad hoc Mode

- If nodes move out of range of the access point (Infrastructure Mode)
- OR nodes are not in direct range of each other (Ad Hoc Mode)
- Then communication is not possible!!

Wireless networks

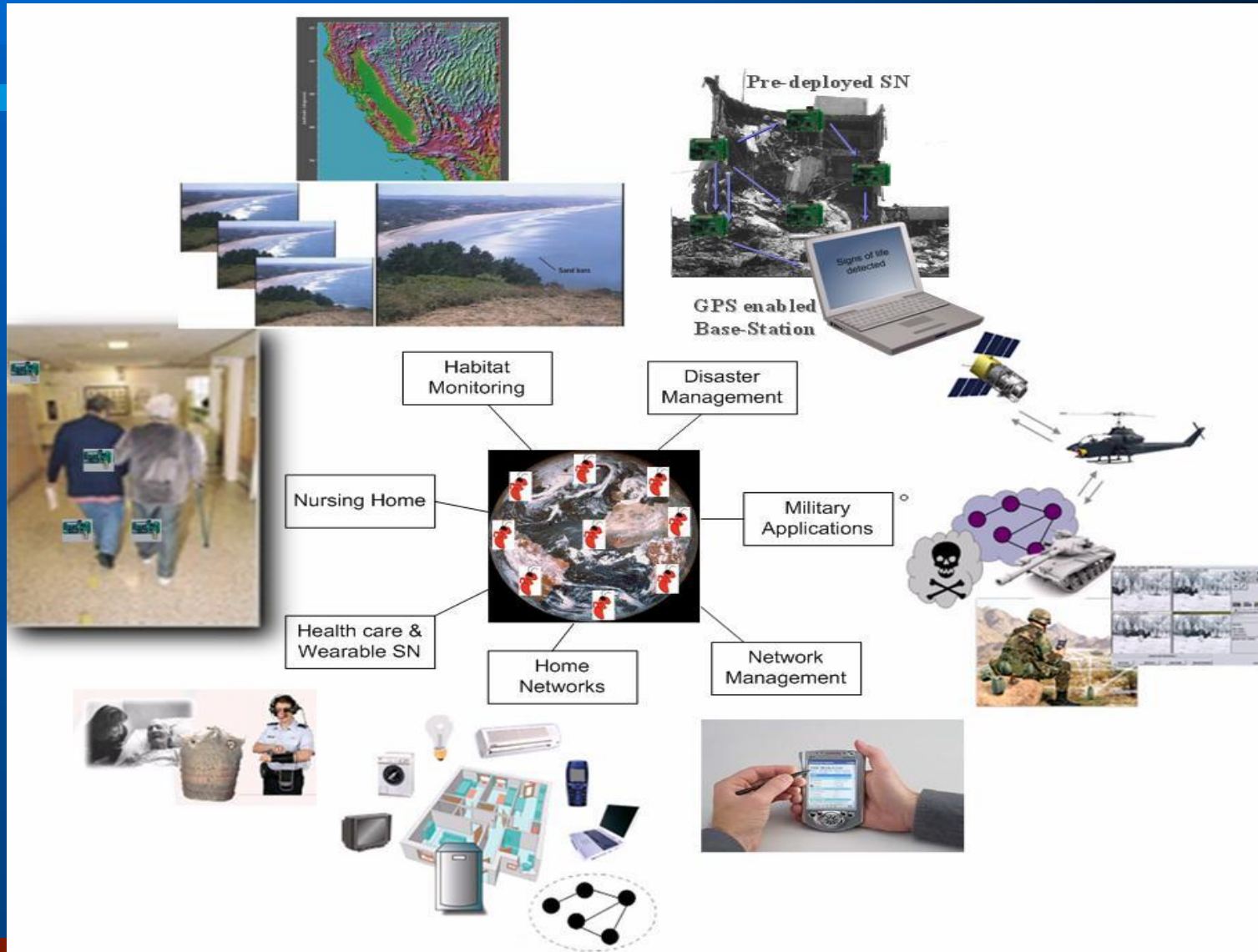


- Wireless LAN, ad-hoc network, wireless mesh, sensor network

AdHoc and Sensors ...

- Ad Hoc and Sensor Networks are both multi-hop wireless architectures
 - Thereby shares several technical issues and challenges
- However, key differences exist
 - Energy constraint in sensor networks
 - Traffic models and characteristics
 - Other issues like coverage, fault-tolerance, etc.

Sensor Networking -- Vision



Sensor Hardware

- *Sensor Node* contains
 1. Sensor(s) to convert a energy form to an electrical impulse
 - e.g., to measure temperature
 2. Microprocessor
 3. Communications link e.g., wireless
 4. Power source e.g., battery

IEEE 802.11 c-r

IEEE 802.11 c

- Bridging functionality for data exchange between wireless and wired networks (MAC layer)

IEEE 802.11 d

- Specification for operation in additional regulatory domains

IEEE 802.11e

- Quality-of-Service support (ongoing work)
- Different service classes, traffic types...

IEEE 802.11f

- Specification of an Inter-Access Point Protocol (IAPP)
- Seamless handover on link-layer; support of different vendors in larger WLANs

IEEE 802.11h (2003)

- Spectrum Management in Europe for 5 GHz band (802.11a)
- Dynamic Frequency Selection/Transmit Power Control

IEEE 802.11i (2004)

- Enhancing Security and Authentication
- Extension of basic WEP (Wired Equivalent Privacy)

IEEE 802.11j

- 4.9-5 GHz adaptation for Japan

IEEE 802.11k

- Enhancements for Radio Resource Measurements (ongoing)

IEEE 802.11n

- Speed enhancements for 802.11
- At least 100 Mbit/s
- Based on MIMO/OFDM up to 540 MBit/s

IEEE 802.11p

- For vehicular usage, speeds up to 200km/h
- 1 km range, 5 GHz frequency band

IEEE 802.11r

- Improves L2 handover, Fast Roaming

IEEE 802.11 s - v

IEEE 802.11s (ongoing)

- Def. of Wireless Distribution Systems and Extended Service Set Mesh Networking
- Define self configuring multi-hop topologies to improve ad-hoc capabilities of 802.11

IEEE 802.11T

- Wireless Performance Prediction (WPP) - test methods and metrics

IEEE 802.11u (ongoing)

- Extension to PHY and MAC layer to improve interoperability with other networks

IEEE 802.11v (ongoing)

- Wireless Network Management
- Extend PHY, MAC in order to support a better radio measurement, better interface to upper layers for managing 802.11 devices in network

IEEE 802.11.2 (ongoing)

- Def. of Performance metrics, measurement methodologies and test conditions

IEEE 802.11REVma (ongoing)

- Task group m, define a Revision 200x of 802.11 incorporating all extensions until now/then.

Study groups

- **APF** Access Point Functionality
- **WIEN** Wireless Interworking with External Networks